

Classification Methodology

Rule-based enrichment, hybrid evaluation, and AI extension

1. Strategy

Both deterministic rules and a local LLM approach were evaluated. The final full-dataset workflow uses deterministic rules because the initial rule coverage was low and full local-LLM execution was not practical in the development environment.

2. Initial Rule Coverage

Path	Tickets	Share
Confident low	3,742	7.8%
Confident security	317	0.7%
Confident outage	8	~0.0%
Rule-classifiable total	4,067	8.5%
Ambiguous / LLM-needed	43,770	91.5%

3. Final Rules-Only Method

- Detect textual evidence associated with severity and operational risk.
- Assign Low, Medium, or Critical priority.
- Generate urgency score and reason.
- Generate boolean operational-risk indicators.
- Record classification status and method.

4. Final Results

Priority	Tickets	Share
Low	45,911	95.97%
Medium	1,605	3.36%
Critical	321	0.67%

- 47,837 tickets successfully enriched.
- 0 failures.
- Approximately 1 minute 47 seconds.

5. AI Extension

Ollama with Qwen2.5 3B was evaluated as a local AI extension. The AI scripts remain as an experimental path for future ambiguity handling. The final dataset is not represented as fully LLM-generated.

6. Limitations

- Rules may miss nuanced context or implicit urgency.
- Production use should include human-labelled validation.
- Urgency is an engineered analytical score, not an SLA metric.